

Discrete Random Variables

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Outline

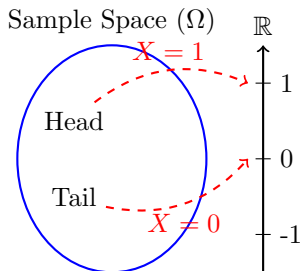
- 1 Random Variables
- 2 Probability Mass Function (PMF)
- 3 Expectation and Variance
- 4 Key Discrete Distributions
- 5 Quiz

Chapter 1: Random Variables

Introduction to Random Variables

What is a Random Variable (RV)?

A **Random Variable** is a function that maps the outcome of a random process (uncertain event) to a real number. It translates “events” into “data”.



Types of Random Variables

Discrete Random Variable

- **Definition:** Takes on a countable number of distinct values (“jumps”).
- **Daily Life Example:** Number of students present in a class (you can’t have 2.5 students).
- **Industry Example:** Number of defective items in a batch of 100 manufactured parts.

Continuous Random Variable

- **Definition:** Takes on an infinite number of values within an interval (“smooth”).
- **Daily Life Example:** The exact time it takes to commute to work (e.g., 25.432 minutes).
- **Industry Example:** The exact weight of a package being shipped (e.g., 10.54 kg).

Numerical Example 1: The “Dice Sum”

Scenario: You roll two fair 6-sided dice. Let X be the random variable representing the **sum** of the two dice.

Task: List the possible values for X .

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Task: List the possible values for X .

- Minimum sum: $1 + 1 = 2$
- Maximum sum: $6 + 6 = 12$
- Range of X : $\{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

Numerical Example 2: Defective Widgets

Scenario: A factory produces light bulbs. You pick 3 bulbs at random to test. Let X be the **number of defective bulbs** found.

Task: List possible values.

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Scenario: A factory produces light bulbs. You pick 3 bulbs at random to test. Let X be the **number of defective bulbs** found.

Task: List possible values.

- All bulbs work $\rightarrow X = 0$
- One is broken $\rightarrow X = 1$
- Two are broken $\rightarrow X = 2$
- All three broken $\rightarrow X = 3$

Sample Space Mapping: $\{GGG\} \rightarrow 0$, $\{BGG, GBG, GGB\} \rightarrow 1$, etc.

Numerical Example 3: Customer Queue

Scenario: A coffee shop opens at 8:00 AM. Let X be the number of customers standing in line exactly at 8:05 AM.

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Scenario: A coffee shop opens at 8:00 AM. Let X be the number of customers standing in line exactly at 8:05 AM.

Task: List possible values.

- No one is there: 0
- 1 person: 1
- ...
- N people: N

Note: Even though there is a physical limit, theoretically, we model this as a set of non-negative integers $\{0, 1, 2, \dots\}$.

Chapter 2: Probability Mass Function (PMF)

Probability Mass Function (PMF)

The PMF, denoted as $P(X = x)$ or $p_X(x)$, gives the probability that the discrete random variable X is exactly equal to some value x .

Properties

- ① $0 \leq P(X = x) \leq 1$ for all x .
- ② $\sum_x P(X = x) = 1$ (The sum of all probabilities must be 1).

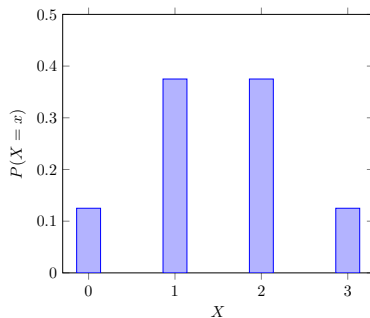
PMF Example 1: Load Balancing

Scenario: A server receives 3 requests. Each has a 50% chance of being “High Priority”. Let X = Number of High Priority requests.

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x	$P(X = x)$
0	$1/8 = 0.125$
1	$3/8 = 0.375$
2	$3/8 = 0.375$
3	$1/8 = 0.125$



PMF Example 2: The Biased Coin

Scenario: You toss a biased coin 3 times. The probability of Heads (H) is 0.6, Tails (T) is 0.4. Let $X = \text{Number of Heads}$.

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Calculations:

- $P(X = 0)$ (TTT): $0.4 \times 0.4 \times 0.4 = 0.064$
- $P(X = 3)$ (HHH): $0.6 \times 0.6 \times 0.6 = 0.216$
- $P(X = 1)$ (HTT, THT, TTH): $3 \times (0.6 \times 0.4 \times 0.4) = 0.288$
- $P(X = 2)$ (HHT, HTH, THH): $3 \times (0.6 \times 0.6 \times 0.4) = 0.432$

Check sum: $0.064 + 0.288 + 0.432 + 0.216 = 1.0 \checkmark$

PMF Example 3: Error Count

Scenario: A coder writes a function. The probability of having x bugs is given by $P(X = x) = k/2^x$ for $x \in \{1, 2, 3\}$, otherwise 0.

Find Constant k:

$$\sum P(X = x) = 1 \implies \frac{k}{2} + \frac{k}{4} + \frac{k}{8} = 1$$
$$\frac{4k + 2k + k}{8} = 1 \implies \frac{7k}{8} = 1 \implies k = 8/7$$

PMF:

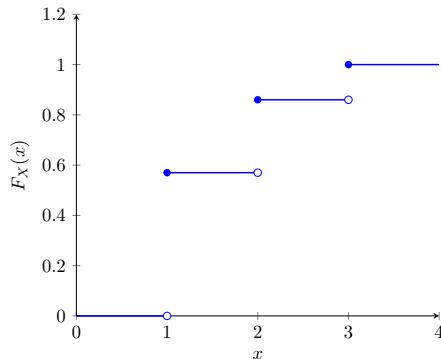
$$P(X = 1) = 4/7, \quad P(X = 2) = 2/7, \quad P(X = 3) = 1/7$$

Cumulative Distribution Function (CDF)

The CDF, $F_X(x) = P(X \leq x)$, represents the running total of probabilities. For discrete variables, it is a **Step Function**.

Using **Example 3 (Errors)**:

- $x < 1$: 0
- $1 \leq x < 2$: $4/7 \approx 0.57$
- $2 \leq x < 3$: $4/7 + 2/7 = 6/7 \approx 0.86$
- $x \geq 3$: 1.0



Chapter 3: Expectation and Variance

Expectation (The Mean)

Definition

The Expectation $E[X]$ is the **weighted average** of the possible values.

$$E[X] = \sum_x x \cdot P(X = x)$$

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Example 1: Fair Die Roll ($X \in \{1, 2, 3, 4, 5, 6\}$)

$$E[X] = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + \cdots + 6\left(\frac{1}{6}\right)$$

$$E[X] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6} = 3.5$$

Expectation: Linearity Property

Property: $E[aX + b] = aE[X] + b$.

Numerical Example: A salesman sells X cars per week. $E[X] = 2$ cars. His salary is defined as: \$500 (Base) + \$200 per car. Let Salary $Y = 200X + 500$.

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Step 1: Apply Linearity

$$E[Y] = E[200X + 500] = 200 \cdot E[X] + 500$$

Expectation: Linearity Property

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Numerical Example: A salesman sells X cars per week. $E[X] = 2$ cars. His salary is defined as: \$500 (Base) + \$200 per car. Let Salary $Y = 200X + 500$.

Step 1: Apply Linearity

$$E[Y] = E[200X + 500] = 200 \cdot E[X] + 500$$

Step 2: Substitute $E[X]$

$$E[Y] = 200(2) + 500 = 400 + 500 = \$900$$

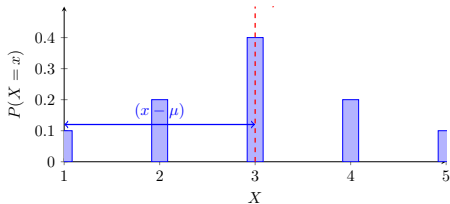
Conclusion: We do not need the full distribution of X to find the average salary, just the average of X .

Variance (Uncertainty) and Standard Deviation

Definition

Variance measures how spread out the discrete data points are from the mean. Let $\mu = E[X]$

$$Var(X) = E[(X - \mu)^2] = \sum (x - \mu)^2 P(X = x)$$



Definition

Standard deviation is the square root of the variance.

$$std(X) = \sigma_X = \sqrt{Var(X)}$$

Alternative Variance Formula

We use the property $Var(X) = E[X^2] - (E[X])^2$ for easier calculation. Let's prove it.

- ① Let $\mu = E[X]$. By definition:

$$Var(X) = E[(X - \mu)^2]$$

- ② Expand the squared term:

$$(X - \mu)^2 = X^2 - 2X\mu + \mu^2$$

- ③ Apply Linearity of Expectation:

$$E[X^2 - 2X\mu + \mu^2] = E[X^2] - 2\mu E[X] + \mu^2$$

- ④ Substitute $E[X] = \mu$:

$$\begin{aligned} &= E[X^2] - 2\mu(\mu) + \mu^2 = E[X^2] - 2\mu^2 + \mu^2 \\ &= E[X^2] - \mu^2 \implies E[X^2] - (E[X])^2 \end{aligned}$$

Variance Property: Scaling and Shifting

Property: If $Y = aX + b$, then $Var(Y) = a^2 Var(X)$.

Proof:

- Let $\mu_X = E[X]$. Mean of Y is $\mu_Y = a\mu_X + b$.
- $Var(Y) = E[(Y - \mu_Y)^2] = E[((aX + b) - (a\mu_X + b))^2]$
- Simplify inside: $(aX - a\mu_X) = a(X - \mu_X)$.
- Square it: $a^2(X - \mu_X)^2$.
- Apply Expectation: $E[a^2(X - \mu_X)^2] = a^2 E[(X - \mu_X)^2] = a^2 Var(X)$.

Note: Adding a constant b simply shifts the distribution, it does not change the spread.

Numerical Example: Scaling Variance

Scenario: Let X be the temperature in Celsius with $Var(X) = 2$. We convert this to Fahrenheit using $Y = 1.8X + 32$.

Question: What is the variance of the temperature in Fahrenheit (Y)?

Numerical Example: Scaling Variance

Scenario: Let X be the temperature in Celsius with $Var(X) = 2$. We convert this to Fahrenheit using $Y = 1.8X + 32$.

Question: What is the variance of the temperature in Fahrenheit (Y)?

Solution: Using the property $Var(aX + b) = a^2 Var(X)$ where $a = 1.8$ and $b = 32$:

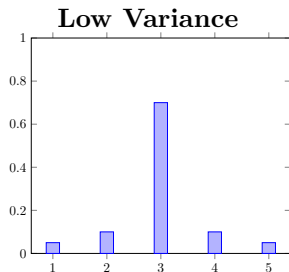
$$Var(Y) = (1.8)^2 \cdot Var(X)$$

$$Var(Y) = 3.24 \cdot 2 = 6.48$$

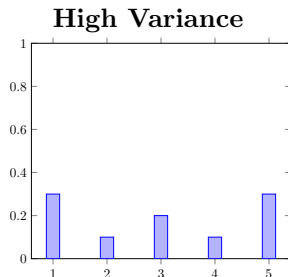
The spread amplifies by the square of the scaling factor.

High vs. Low Variance (Same Mean)

Both distributions below have the same Mean ($\mu = 3$), but different Variances.



Data is clustered near 3.



Data is spread far from 3.

Numerical Example: Calculating Variance

Let X be the number of daily complaints.

x	0	1	2
$P(X = x)$	0.2	0.5	0.3

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Step 1: Mean $\mu = E[X]$

$$\mu = 0(0.2) + 1(0.5) + 2(0.3) = 0 + 0.5 + 0.6 = 1.1$$

Numerical Example: Calculating Variance

Let X be the number of daily complaints.

x	0	1	2
$P(X = x)$	0.2	0.5	0.3

Step 1: Mean $\mu = E[X]$

$$\mu = 0(0.2) + 1(0.5) + 2(0.3) = 0 + 0.5 + 0.6 = 1.1$$

Step 2: Variance $E[(X - \mu)^2]$

- $x = 0 : (0 - 1.1)^2 \times 0.2 = 1.21 \times 0.2 = 0.242$
- $x = 1 : (1 - 1.1)^2 \times 0.5 = 0.01 \times 0.5 = 0.005$
- $x = 2 : (2 - 1.1)^2 \times 0.3 = 0.81 \times 0.3 = 0.243$

$$\text{Var}(X) = 0.242 + 0.005 + 0.243 = 0.49$$

Chapter 4: Different Discrete Random Variables and Distributions

1. Discrete Uniform Distribution

Definition: The random variable takes any integer value between a and b with equal probability.

Notation: $X \sim U(a, b)$.

Example: Die Roll ($a = 1, b = 6$)

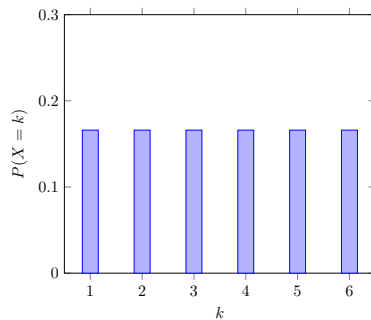
- $n = 6$, Mean = 3.5

Parameter: $n = b - a + 1$ (Total outcomes)

PMF: $P(X = k) = \frac{1}{n}$ for $k \in \{a, a + 1, \dots, b\}$

Properties:

- **Mean:** $E[X] = \frac{a+b}{2}$
- **Variance:** $Var(X) = \frac{(b-a+1)^2-1}{12} = \frac{n^2-1}{12}$



2. Bernoulli Distribution

Definition: Single trial with binary outcome (0 or 1).

Use Cases:

- Tossing a coin
- Pass/Fail in an exam.
- an item being defective or not.

Notation: $X \sim \text{Bern}(p)$

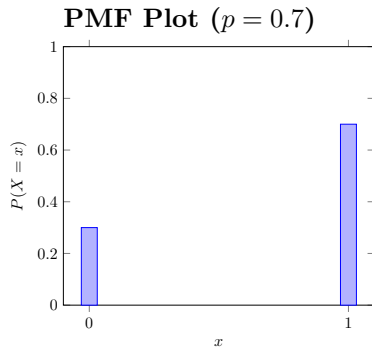
Parameter: p (prob of success).

PMF:

$$P(X = x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases}$$

Properties:

- **Mean:** $E[X] = p$
- **Variance:** $\text{Var}(X) = p(1 - p)$



3. Binomial Distribution

Definition: Number of successes in n independent Bernoulli trials.

Use Cases:

- **Quality Control:** If you test 20 items (n), and each has 10% chance of defect (p), what is the probability that less than 5 items are defective?

Notation: $X \sim \text{Bin}(n, p)$

Parameters:

- n : The total number of independent trials.
- p : The probability of success in a single trial.

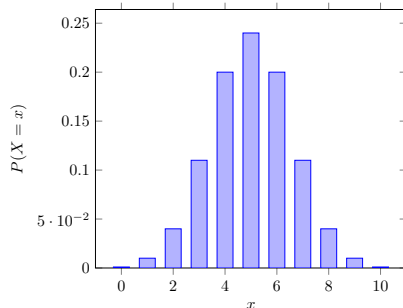
PMF:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \quad ; \quad 0 \leq k \leq n$$

Properties:

- **Mean:** $E[X] = np$
- **Variance:** $\text{Var}(X) = np(1 - p)$

PMF ($n = 10, p = 0.5$)



4. Poisson Distribution

Definition: Models the number of events occurring in a fixed interval of time or space.

Use Cases:

- **Queues:** the probability of receiving a specific number of customer service calls during a peak hour.
- **Website Traffic:** Predicting the number of visitors or server requests per minute.

Notation: $X \sim Pois(\lambda)$

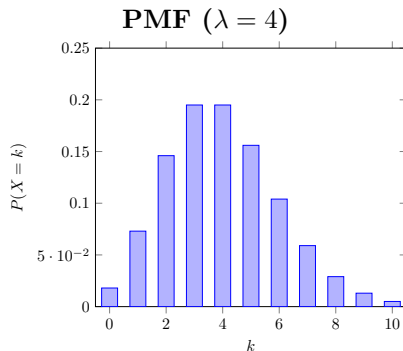
Parameter: λ : The average number of events per interval.

PMF:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Properties:

- **Mean:** $E[X] = \lambda$
- **Variance:** $Var(X) = \lambda$



Chapter 5: Quiz

Quiz: Question 1

Q1: You are training a model. You have a random variable X with $Var(X) = 10$. You decide to normalize the data by doubling it and adding 5 ($Y = 2X + 5$). What is the variance of the new variable Y ?

- A) 25 B) 20 C) 40 D) 45

Quiz: Question 1

Q1: You are training a model. You have a random variable X with $Var(X) = 10$. You decide to normalize the data by doubling it and adding 5 ($Y = 2X + 5$). What is the variance of the new variable Y ?

A) 25 B) 20 C) 40 D) 45

Answer: C) 40

$$Var(aX + b) = a^2 Var(X)$$

$$Var(2X + 5) = 2^2 \cdot 10 = 4 \cdot 10 = 40$$

Quiz: Question 2

Q2: A call center receives an average of $\lambda = 3$ calls per minute (Poisson). What is the probability of receiving exactly 0 calls in a minute?

- A) 0 B) e^{-3} C) $3e^{-3}$ D) 0.5

Quiz: Question 2

Q2: A call center receives an average of $\lambda = 3$ calls per minute (Poisson). What is the probability of receiving exactly 0 calls in a minute?

A) 0 B) e^{-3} C) $3e^{-3}$ D) 0.5

Answer: B) e^{-3}

$$P(X = 0) = \frac{e^{-3} \cdot 3^0}{0!} = e^{-3}$$

Quiz: Question 3

Q3: If $E[X] = 5$ and $E[X^2] = 30$, what is the Standard Deviation σ ?

- A) 5 B) $\sqrt{5}$ C) 25 D) $\sqrt{30}$

Quiz: Question 3

Q3: If $E[X] = 5$ and $E[X^2] = 30$, what is the Standard Deviation σ ?

A) 5 B) $\sqrt{5}$ C) 25 D) $\sqrt{30}$

Answer: B) $\sqrt{5}$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 30 - 25 = 5 \implies \sigma = \sqrt{5}$$

Quiz: Question 4

Q4: We perform 5 independent trials. Each trial has a success probability of 0.2. Which distribution models the number of successes?

- A) Bernoulli(0.2)
- B) Poisson(5)
- C) Binomial(5, 0.2)
- D) Uniform

Quiz: Question 4

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- A) Bernoulli(0.2)
- B) Poisson(5)
- C) Binomial(5, 0.2)
- D) Uniform

Answer: C) Binomial(5, 0.2) Because we are counting successes in a fixed number ($n = 5$) of independent trials.

Quiz: Question 5

Q5: Which of the following variables is Discrete?

- A) The time it takes for a website to load.
- B) The number of clicks on an advertisement.
- C) The temperature of the server room.
- D) The height of a user.

Quiz: Question 5

Q5: Which of the following variables is Discrete?

- A) The time it takes for a website to load.
- B) The number of clicks on an advertisement.
- C) The temperature of the server room.
- D) The height of a user.

Answer: B) The number of clicks Clicks are countable integers (0, 1, 2...). Time, temperature, and height are continuous.

- **Random Variables** map real-world uncertainty to numbers.
- **PMF** gives prob of specific values; **CDF** gives prob of “less than or equal”.
- **Expectation** is the center; **Variance** is the spread.
- **Key Distributions:** Bernoulli (1 trial), Binomial (n trials), Poisson (Rates/Queues).

Next Step: Continuous Random Variables & Normal Distribution.